

PROCESS FUNDAMENTALS OF BASIC RESEARCH

I. Research Problem

Approach

Qualitative

Quantitative

1. Define the problem

What is the purpose of the study? What information is needed? How can we operationalize the terms? How will the information be used?

2. Review previous research

What studies have already been done on this topic? Do we need additional information before we begin? From what perspective should we approach the issue?

3. Determine the research design

Can we use existing data? What will we measure or observe? What research methods should we use?

2. Research Design

To find out how people behave in public

Watch them

Field

To find out how people behave in private

Ask them to keep diaries, video, audio, graphic, symbolic, etc.

Document

To learn what people think

Ask them

Survey/Interview

To appreciate before/after, beginning/ending, success/challenges

Administer evaluation instruments

Evaluation

To identify trends in verbal material

Systematic tabulation

Content Analysis

To understand an unusual event

Detailed and lengthy investigation

Case

Sample/census; collect data

Are we interested in a specific population? How large should the sample be? Who will gather the data? How long will it take?

3. Sampling/Census

Simple Random

Assign to each population a unique number; select sample items by use of random numbers

Systematic

Use natural ordering or order population; select random starting point between 1 and the nearest integer to the sampling ratio (N/n); select items at interval of nearest integer to sampling ratio

Multistage Random

Use form of random sampling in each of the sampling stages where there are at least two stages

Quota

Classify population by pertinent properties; determine desired proportion of sample from each class; fix quotas for each observer

Data Collection

Data collection instruments

Pre coded for data entry into a computer

Graphic displays

Bar charts, histograms and frequency polygons

Frequency distribution

Displays variation in a form that can be easily inspected and described

Summary statistics

Describes the central tendency and variability of distributions

Cross tabulations

Percentaged within the categories of independent variables

Crossbar analysis

Tests for spurious and mediating relationships to identify why they occur

Inferential statistics

Estimates the confidence placed in statistical estimate

5. Statistical Analysis

Frequency

Proportion, Percent: the relative frequency of occurrence for each variable value

Frequency: frequency of occurrence for each variable value

Central Tendency

Mean: indicator of central tendency sensitive to the exact position of each score in the distribution

Mode: most commonly occurring value

Median: middle source in a distribution

Dispersion/Variability

Range: difference between the lowest and highest scores

Standard Deviation: indicator of average deviation from mean

Variance: average squared deviation from mean

6. Complete analysis/draw conclusions

What statistical techniques will we use?

7. Prepare research report

Who will read the report? What is their level of familiarity with the subject? How should we construct the report?